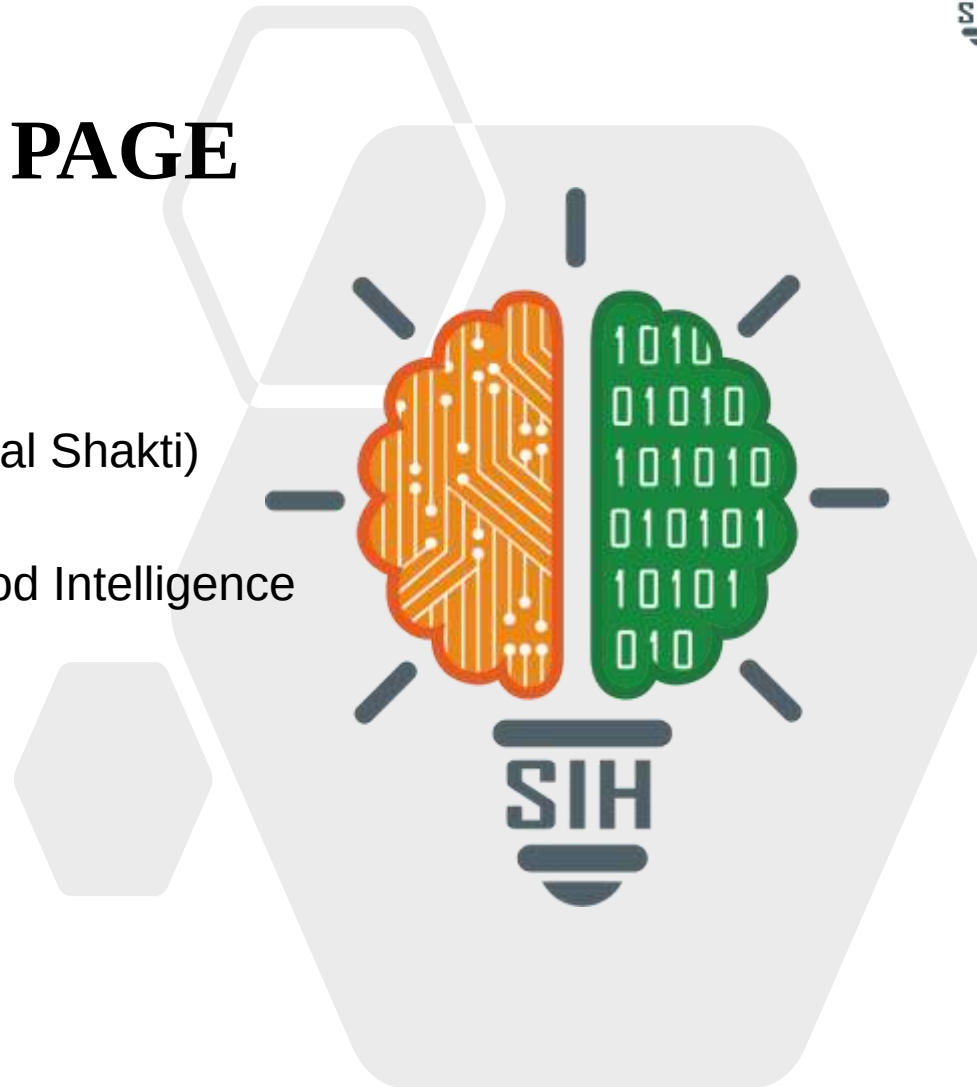


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID-**
Problem Statement ID - SIH1234 (Ministry of Jal Shakti)
- **Problem Statement Title-**
Problem Statement Title - Real-time Flash Flood Intelligence
- **Theme-**
Theme - Disaster Management
- **PS Category- Software/Hardware**
PS Category- Software
- **Team ID-**
Team ID - 123456
- **Team Name (Registered on portal)**
Team Name - Code Mavericks



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- Detailed explanation of the proposed solution
A military-grade, real-time command center that fuses predictive LSTM hydrological AI with a zero-latency Telegram dispatch protocol. It visualizes threats via an Interactive Radar.
- How it addresses the problem
 - Eliminates catastrophic communication latency between monitoring stations (CWC) and ground-level response forces (NDRF) during flash floods.
- Innovation and uniqueness of the solution
 - Bypasses traditional, slow SMS gateways by integrating directly with Telegram API for instant, encrypted mass-broadcasting. Includes a frontend-fallback architecture.

TECHNICAL APPROACH

- Technologies to be used (e.g. programming languages, frameworks, hardware)
 - Frontend: React, Tailwind CSS (Glassmorphism UI)
 - Backend: Python FastAPI, Custom Hydro-LSTM AI Engine
 - Integration: Telegram Bot API (Direct Dispatch), Vercel
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

Methodology & User Flow:

1. Ingestion: IoT Sensors & CWC data feed into Python LSTM.
2. Visualization: React frontend displays live sweeping Radar.
3. Triage: Commander clicks 'DISPATCH ALERTS'.
4. Broadcast: Zero-latency API hits NDRF Chat IDs instantly.



Your
Team
Name

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FEASIBILITY AND VIABILITY



- Analysis of the feasibility of the idea
- Potential challenges and risks
 - Highly Feasible (Business to Government) SaaS Model.
 - Extreme cost-efficiency using Serverless Edge architectures and free-tier Telegram API for massive broadcast capabilities.
- Strategies for overcoming these challenges
 - Potential Challenges: Backend server failure or DNS routing issues during a disaster. Poor internet (2G) in remote basins.
 - Strategies: Dashboard uses 'Frontend-Fallback' to bypass broken backends and dispatches directly from the browser to Telegram. Telegram API uses minimal data (KB), succeeding on 2G networks.

IMPACT AND BENEFITS

- Potential impact on the target audience
- Target Audience Impact: Empowers District Magistrates and NDRF commanders with instantaneous, actionable intelligence, reducing decision fatigue and cutting evacuation response times by 80%.
(social, economic, environmental, etc.)
- Social Benefit: Rapid, targeted evacuations in high-risk zones (e.g., Alaknanda Fault) directly preventing loss of life.
- Economic & Technical Benefit: Early warnings secure infrastructure. Establishes a modernized, secure Indian disaster communication protocol.



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RESEARCH AND REFERENCES



- **Details / Links of the reference and research work**
 - Ministry of Jal Shakti Guidelines on Flood Forecasting.
 - Official Telegram Bot API Documentation for high-throughput messaging.
 - Advanced LSTM neural networks for hydrological time-series prediction.
 - Defense and aviation sector UX research for low-clutter interfaces.